Cold Chain Storage and Transport - Standard Operating Procedure

Document HPD-SOP-002, version 1.4, effective 2026-02-15.

1. Purpose

This procedure describes how Helix stores and transports temperature-sensitive medicines so that they reach customers within their labelled storage conditions. It applies to warehouse staff, drivers and the Quality Assurance (QA) team.

2. Storage temperature ranges

- Refrigerated products (vaccines, insulin, biologics): 2 to 8 °C.
- Controlled room temperature products: 15 to 25 °C.
- Frozen products (-20 °C or colder) are not stocked by Helix.

3. Temperature monitoring

Every cold room, refrigerator and transport box is monitored by a calibrated data logger that records the temperature every 5 minutes. An alarm is raised when the temperature stays outside the allowed range for more than 15 minutes. A QA pharmacist reviews the logger data every working day. Temperature records are kept for 5 years. Data loggers are calibrated once a year by an accredited laboratory.

4. Handling a temperature excursion

An excursion is any period during which a product is outside its labelled range.

- If a 2 to 8 °C product was above 8 °C for 30 minutes or less in total, it may stay in stock. The excursion must be documented in the excursion log.
- If the excursion lasted more than 30 minutes in total, or the product was below 0 °C for any length of time, the product is quarantined immediately and labelled DO NOT USE.
- QA must be notified within 1 hour of a quarantine. QA decides, using the manufacturer's stability data, whether the product can be released or must be destroyed. The decision is made within 2 business days.

5. Transport

Refrigerated products travel in validated insulated boxes with conditioned ice packs, or in refrigerated vehicles. The maximum transit time is 8 hours for deliveries inside Tehran and 24 hours for inter-city deliveries; inter-city deliveries of refrigerated products always use refrigerated vehicles. The driver checks the data logger at hand-over and the customer signs the delivery note only after seeing that no alarm was raised.

6. Power failure

Cold rooms are connected to a backup generator that must start within 10 minutes of a power failure. If a cold room is expected to stay above 8 °C for more than 2 hours, stock is moved to the backup cold room and the move is recorded.